

1. Program for Linear Search

```
#include<stdio.h>
```

```
int main() {
```

```
    // declaration of the array and other variables
```

```
    int arr[20], size, key, i, index;
```

```
    printf("Number of elements in the array: ");
```

```
    scanf("%d", &size);
```

```
    printf("Enter elements of the array: ");
```

```
    // loop for the input of elements from 0 to number of elements-1
```

```
    for (i = 0; i < size; i++)
```

```
        scanf("%d", &arr[i]);
```

```
    printf("Enter the element to search ie. key element: ");
```

```
    scanf("%d", &key);
```

```
    // loop for traversing the array from 0 to the number of elements-1
```

```
    for (index = 0; index < size; index++)
```

```
        if (arr[index] == key) // comparing each element with the key element
```

```
            break; // cursor out of the loop when a key element found
```

```
    if (index < size) // condition to check whether previous loop partially traversed  
or not
```

```
        printf("Key element found at index %d", index); // printing the index if key  
found
```

```
    else
```

```
        printf("Key element not found");
```

```
    return 0;
}
```

Output:

Number of elements in the array: 6

Enter elements of the array: 12

23

45

67

78

89

Enter the element to search ie. key element: 67

Key element found at index 3

2. Program for Linear search

Occurrences of same number in multiple indexes

```
#include<stdio.h>

int main() {
    // declaration of the array and other variables
    int arr[20], size, key, i, index;
    int countKey = 0; //initializing count of key element as 0
    printf("Number of elements in the array: ");
    scanf("%d", &size);
```

```

printf("Enter elements of the array: ");
// loop for the input of elements from 0 to number of elements-1
for (i = 0; i < size; i++)
    scanf("%d", &arr[i]);

printf("Enter the element to search ie. key element: ");
scanf("%d", &key);

// loop for traversing the array from 0 to the number of elements-1
for (index = 0; index < size; index++) {
    if (arr[index] == key) { // comparing each element with the key element
        printf("Key element found at index %d\n", index); // printing the index if
key found
        countKey++; // incrementing the count of key element;
    }
}

if (countKey == 0) // condition to check whether key element is found or not
    printf("Key element not found");
else
    printf("Key element is present %d times in the array.\n", countKey);

return 0;
}

```

Output

Number of elements in the list: 5

Enter elements of the list: 5

5

5

5

5

Enter the element to search ie. key element: 5

Key element found at index 0

Key element found at index 1

Key element found at index 2

Key element found at index 3

Key element found at index 4

Key element is present 5 times in the array.

3. Program for binary search

// Binary Search in C

```
#include <stdio.h>
```

```
int binarySearch(int array[], int x, int low, int high) {
```

```
    // Repeat until the pointers low and high meet each other
```

```
    while (low <= high) {
```

```
        int mid = low + (high - low) / 2;
```

```
        if (array[mid] == x)
```

```
            return mid;
```

```
        if (array[mid] < x)
```

```
            low = mid + 1;
```

```
        else
```

```
            high = mid - 1;
```

```
    }
```

```
    return -1;
```

```
}
```

```
int main(void) {  
    int array[] = {3, 4, 5, 6, 7, 8, 9};  
    int n = sizeof(array) / sizeof(array[0]);  
    int x = 9;  
    int result = binarySearch(array, x, 0, n - 1);  
    if (result == -1)  
        printf("Not found");  
    else  
        printf("Element is found at index %d", result);  
    return 0;  
}
```

Output

Element is found at index 6